

Physical Storage & Indexing Strategies.

Lab 1: Inspecting Table Size

Query

Query History

```

1  SELECT
2      relname AS object_name,
3      pg_size_pretty(pg_table_size(fed.oid)) AS data_size,
4      pg_size_pretty(pg_indexes_size(fed.oid)) AS index_size,
5      pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size
6  FROM pg_class fed
7  JOIN pg_namespace n ON n.oid = fed.relnamespace
8  WHERE fed.relkind = 'r' AND n.nspname = 'public'
9  ORDER BY pg_total_relation_size(fed.oid) DESC;

```

Data Output

Messages

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SQL

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	object_name name	data_size text	index_size text	total_size text
1	rental	1232 kB	1120 kB	2352 kB
2	payment	896 kB	920 kB	1816 kB
3	film	736 kB	200 kB	936 kB
4	film_actor	272 kB	216 kB	488 kB
5	inventory	232 kB	208 kB	440 kB
6	customer	96 kB	112 kB	208 kB
7	address	88 kB	64 kB	152 kB
8	city	64 kB	48 kB	112 kB
9	film_category	72 kB	40 kB	112 kB
10	actor	40 kB	32 kB	72 kB
11	store	8192 bytes	32 kB	40 kB
12	staff	16 kB	16 kB	32 kB
13	language	8192 bytes	16 kB	24 kB
14	category	8192 bytes	16 kB	24 kB
15	country	8192 bytes	16 kB	24 kB

Lab 2: The Impact of a New Index

Step 1 : Check current size of 'payment ' indexes

Query

Query History

1 SELECT pg_size_pretty(pg_indexes_size('payment'));

Data Output

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SQL

Showing rows: 1 to 1

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	pg_size_pretty text
1	920 kB

Step 2: Create a heavy index on payment_date

Query Query History



```
1 CREATE INDEX idx_payment_date_long
2 ON payment (payment_date, amount, customer_id);
```

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CREATE INDEX

Query returned successfully in 3 secs 696 msec.

Step 3: Check size again

Query

Query History

1

`SELECT pg_size_pretty(pg_indexes_size('payment'));`

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SQL

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	pg_size_pretty text
1	1488 kB

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Lab 3: Partial Index Practice

1. Create Partial Index

Query

Query History

1

2

3

```
CREATE INDEX idx_expensive_films
ON film (replacement_cost)
WHERE replacement_cost > 25.00;
```

Data Output

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```
CREATE INDEX
```

```
Query returned successfully in 2 secs 828 msec.
```

Find the size of this specific index

Query

Query History

1

`SELECT pg_size_pretty(pg_relation_size('idx_expensive_films'));`

Data Output

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SQL

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	pg_size_pretty text
1	16 kB

2. Full Index

Query Query History



```
1 CREATE INDEX idx_full_replacement_cost
2 ON film (replacement_cost);
```

Data Output Messages Notifications



CREATE INDEX

Query returned successfully in 1 secs 219 msec.

Query Query History



```
1 SELECT
2     'Partial Index' AS index_type,
3     pg_size_pretty(pg_relation_size('idx_expensive_films')) AS siz
4 UNION ALL
5 SELECT
6     'Full Index' AS index_type,
7     pg_size_pretty(pg_relation_size('idx_full_replacement_cost'))
```

Data Output Messages Notifications



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	index_type	size
1	Partial Index	16 kB
2	Full Index	40 kB